

Stats-2

Skewness :-

$d = [1, 2, \underline{100}, \underline{200}, 3, 4, 8, 10, 5, 2, 3, 6]$

- ① Zero skewness (Symmetric) ✗
- ✓ ② Right skewness (Positive Skewness)
- ✓ ③ Left skewness (Negative Skewness)

✓ $\text{mean} = 28.66$

✓ $\text{median} = 4.5$

✓ $\text{mode} = 2, 3$

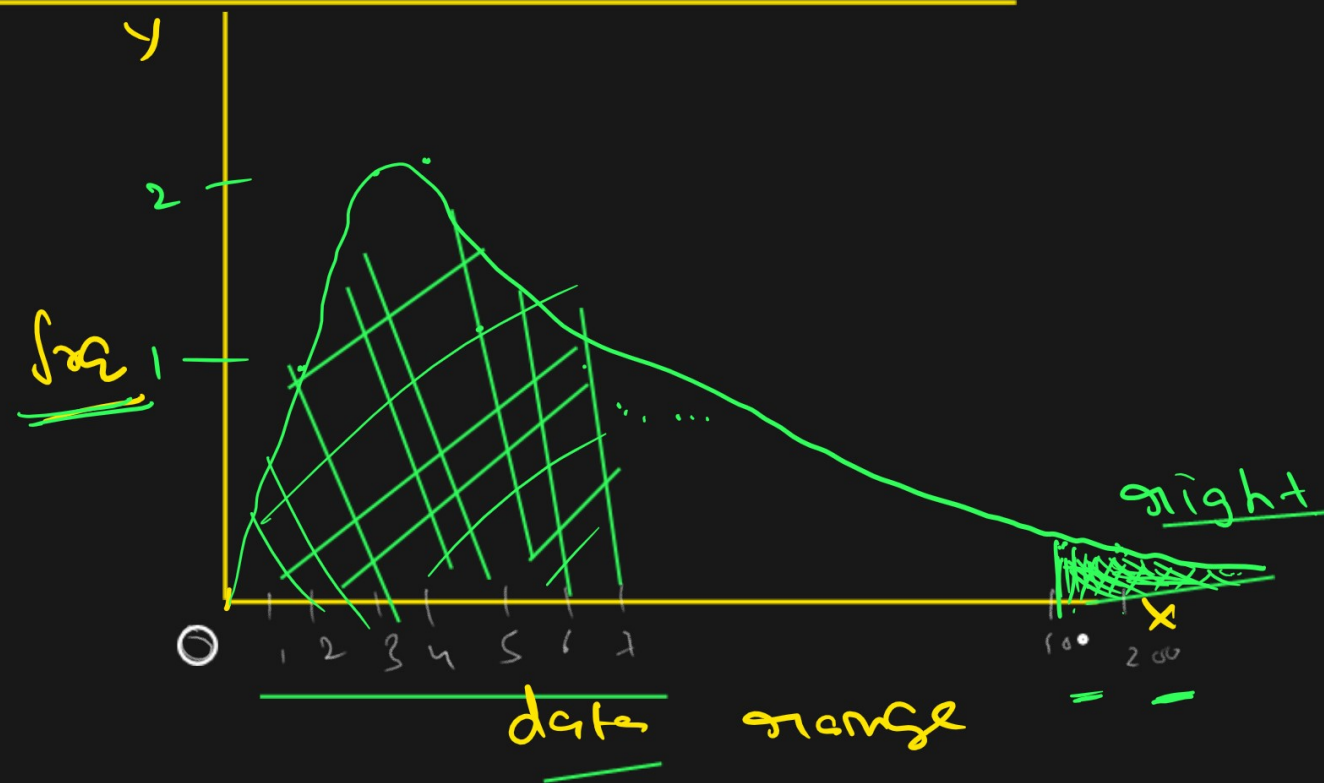
$\text{mean} = \text{median} = \text{mode}$ ✗

$\text{mean} > \text{median} > \text{mode}$ → Positive Skewness

✓ $\text{mean} < \text{median} < \text{mode}$ → Negative Skewness.

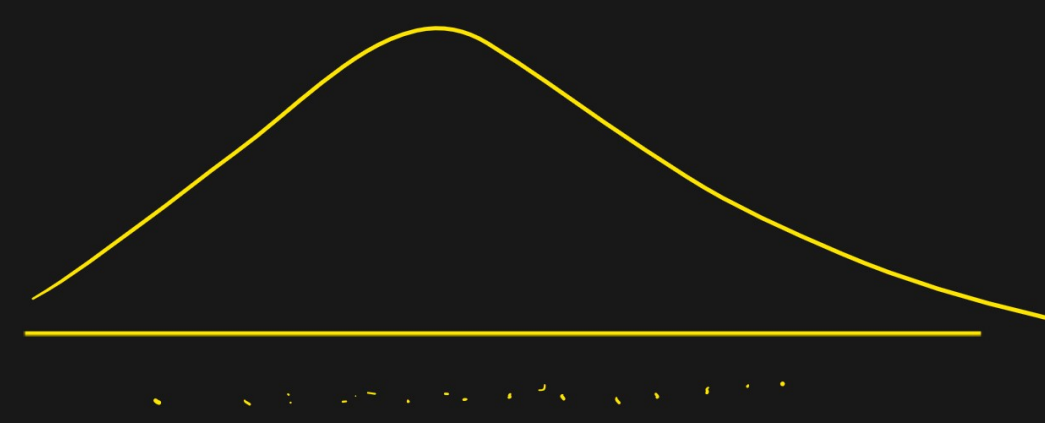
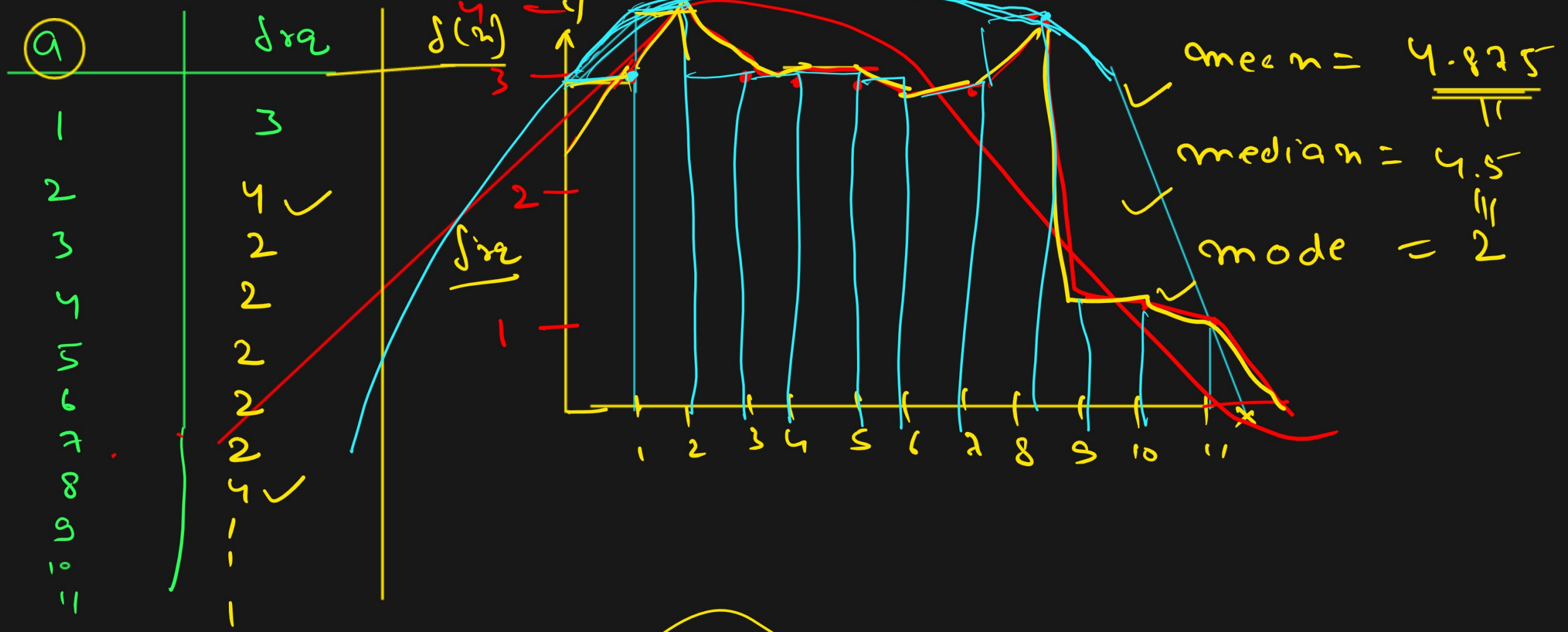
$a = [1, 2, 100, 200, 34, 8, 10, 5, 2, 3, 6]$

(a)	freq
1	1
2	2
100	1
200	1
3	2
4	1
8	1
10	1
5	1
6	1



$a_1 = [1, 2, 3, 2, 4, 1, 5, 6, 7, 8, 10, 11, 1, 2, 3, 4, 8, 5, 5, 8, 7, 6, 8, 2]$





Normal Distribution

Gaussian Distribution

$$\mu = 0 = \text{mean}$$
$$\sigma = 1 = \text{std.}$$

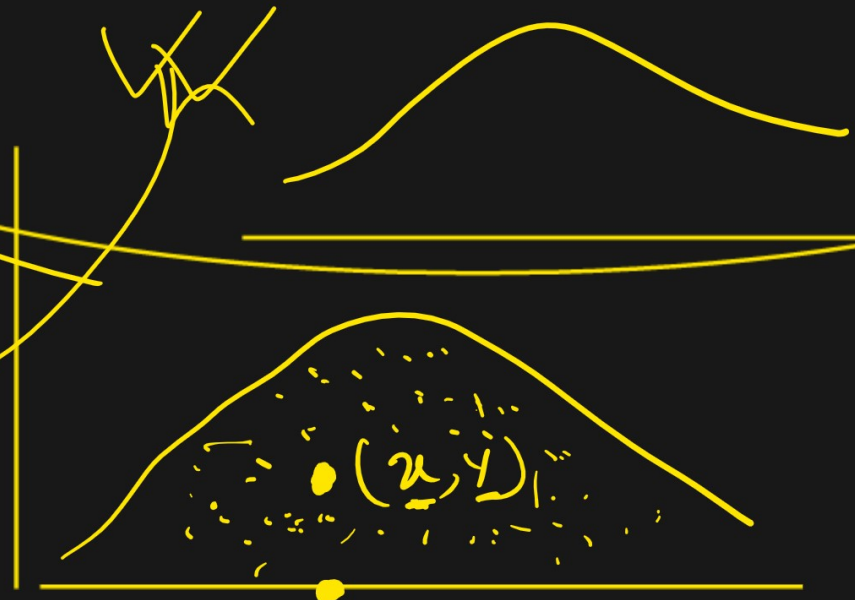
Standard Normal dist

① mean

$$= \mu$$

② Standard deviation

$$= \sigma$$



$$(PDF) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$